System Administration Tasks

CSC252

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Accounts

- Files
  - /etc/passwd – User information
    - ID, group ID, home dir, shell, ...
  - /etc/shadow
    - Encrypted user password
  - NIS, NIS+
    - Network Information Service (Originally called: yellow pages)
    - Allowed for distribution of system configuration data
      - Included user accounts, hostnames.
    - Sun Microsystems developed NIS+ to replace it
      - Completely different design, but achieves the same goal
    - Both have been replaced at this point – The protocols were designed to be replaced

Accounts (cont.)

- Lightweight Directory Access Protocol (LDAP)
  - Distribute information across the network
  - Managed from a central server (Can be multiple for redundancy)
  - RedHat Documentation
- Active Directory
  - Microsoft’s implementation
  - Meant for managing Windows Environments
  - Can be used across platforms

Adding local users

- useradd [-u uid] [-o] [-g group] [-G group [, group, …] [-d home] [-m] [-k template] [-f inactive] [-r] [-p passwd] [-s shell] [-e expire] [-c comment] login

- Default Settings
    - useradd –D [-g group] [-b base] [-f inactive] [-e expire] login
    - useradd –D
        - Look at default settings

User Management

- usermod [-u uid] [-o] [-g group] [-G group (, group, …)] [-d home] [-m] [-l new_login] [-f inactive] [-r] [-p passwd] [-s shell] [-e expire] [-c comment] login

- userdel [-r] login

- passwd <login>
  - Change the password for a specific user (must be root)

System Accounts

- root (superuser)
- daemon (system server processes/files)
- bin (owns system executables and files)
- sys (for Distributed File Services, DFS)
- lp (managing printers)
- adm (basic system functions)
- nobody (default for unprivileged operations)

- Viewing system accounts: https://www.cyberciti.biz/faq/linux-list-users-command/

Groups

- Primary Group
- Secondary Groups
  - /etc/group - Holds the groups available on the system

- groups command
  - get a list of groups
  - user can change current group with **newgrp**

System Groups

- root
- daemon
- sys
- tty
- bin
- adm
- lp
- mail
- uucp/nuucp
- nobody
- noaccess

Backing up

- backup from one machine
- label all tapes
- Three generations
  - Grandparent / Parent / Child
  - 3 Backup cycle (Daily, Weekly, Monthly)
  - avoids propagating an error
- secure backup tapes
- keep tapes off site
- backup appropriately
- Always check your backup procedure
  - Attempt a restore
- Tape life Cycle – How often should tapes be replaced?

Backup Schedule Depends on

- activity on filesystem
- capacity of dump device
- length of “dump window”
- amount of redundancy
- number of tapes available
- how far back you must be able to recover

Linux Device Names

- [Linux Naming Convention]

- /dev directory
- fd0 – Floppy Drives (if you know what those are)
- sda (First SCSI or SATA Hard Disk)
- sdb (Second SCSI or SATA Hard Disk)
- scd0/sr0 – CD-ROM
- mmcblk0 – SDHC/PCMCIA – mvne ssds
- Partitions are represented by appending a decimal number:
  - sda1 and sda2 represent the first and second partitions of the first SCSI disk drive

Disk Information

Get information about available disks:

- df command
  - Disk space usage
- lsblk command
  - List the block devices connected
- lsusb command
- fdisk
  - Used to partition disks

Adding new drives

- Physically add drives
- Create entries in /dev
  - mknod command
- format the drive
- mkfs (make the file system)
- Add entry to /etc/fstab (optional)
  - Allows for mounting devices at start time

Mounting Filesystem

- mount [-r] [device] <directory>
  - uses /etc/fstab
  - -r for read only
- mount –t type [device] <directory>
  - specify the type of device
- mount [-f filesystem_types]
  - "Fake" don't actually mount anything
  - See what mount is trying to do,
  - Add an entries to /etc/mtab
- mountall
  - called during boot process
  - uses /etc/fstab to get disk info

Unmount Filesystems

- umount <filesystem>
- umountall

Filesystem Commands

- Who is using a file a directory or a socket?
  - fuser [-u] [-k] <device>

    -u for user

    -k kills the process using the device

    -v is verbose
- Checking a filesystem
  - fsck

fsck tasks

- Fixes damaged or inconsistent indoes
- unreferenced inodes
- large link counts
- unused data blocks not recorded in block maps
- data blocks listed as free but also used in a file
- incorrect summary info in a superblock

quotas

- Set up user quotas (Place a limit on disk space usage available to user(s))
- rq in /etc/fstab
- create a quotas file in the affected filesystem

- edquotea
  - Allows for editing of quotas
- quota
  - display quotas

Network File Systems - NFS

- Allows access to storage devices over a network
  - Appears to be connected locally
  - Runs on RPC protocol (Remote Procedure Call)

- daemons
  - nfsd, mountd, statd, lockd, nfslogd
  - start when system enters multiuser mode

- /etc/exports

  Specifies exported file systems and hosts permitted to access them

Boot Process

- Loading and initialization of the kernel
- Device detection and configuration
- Creation of system processes
- Operator intervention (manual boot only)
- Execution of startup scripts (by init)
- Multiuser operation (init spawns getty process)

init states

- (Run Levels)

- 0 system is completely shut down
- 1 single-user mode (administration)
- 2 multi-user mode without NFS (networking)
- 3 full multi-user mode with networking
- 4 unused
- 5 full multi-user mode with networking and X11 (GUI)
- 6 reboot

- startup scripts are defined in /etc/rc#.d
- commands in /etc/rc.local will execute on a runlevel change

Startup Script Tasks

- setting the name of the computer
- setting the timezone
- checking the disks with fsck
- mounting the system's disks
- removing old files in /tmp
- configuring network interfaces
- starting up daemons and network services

Shutdown Steps

- logs shutdown
- kills nonessential processes
- executes sync
- waits for filesystem writes to complete
- halts the kernel

Shutdown Commands

- shutdown [TIME] [message]
- halt
- reboot [-p] (-p for poweroff)
- init <run_level>
  - Usually a command ran by the kernel
  - Can be used by root to restart at the given run_level

sync

- flushes cached superblocks to disk
- flushes modified inodes and cached data blocks

Patches

- Updates the system files
- Correct errors or provide security fix

- patch

Software

- CD
- Source Code
- Download Packages
  - deb → tar file with information to install a system
  - System Package Managers

Maintaining Software Packages

- yum
- apt-get

yum

yum install <packageName>	Allows for installing a package and any dependencies (Required Packages)
yum update <packageName>	Update a specific package or all packages
yum check-update	List packages that have updates available to them
yum clean all	Remove all header files used for resolving dependencies and cached packages
yum list available	List all available packages
yum search <word>	Search's for words in a package name, description

Groups of packages

yum grouplist

yum groupinfo

Configuration File: yum.conf

apt-get (apt)

apt update	Update local package list
apt check	
apt install package	
apt remove package	
apt upgrade	Upgrade all packages on the system that don’t require any new packages
apt dist-upgrade	Upgrade all packages on the system and install any new packages necessary

Daemon

- A background process that performs a specific function or system-related task

SYSLOGD – System Logging

- Syslog – Standardized System for managing system log files
  - Programs write to special file (/dev/log)

  - syslogd reads messages from the file
  - consults configuration file syslog.conf
  - dispatches each message to appropriate destination

- sysadmins should back up and maintain log files
- More Information

SYSLOGD – Message Severity Levels

Value	Severity	Keyword	Deprecated keywords	Description	Condition
0	Emergency	`emerg`	`panic` [7]	System is unusable	A panic condition.[8]
1	Alert	`alert`		Action must be taken immediately	A condition that should be corrected immediately, such as a corrupted system database.[8]
2	Critical	`crit`		Critical conditions	Hard device errors.[8]
3	Error	`err`	`error` [7]	Error conditions	
4	Warning	`warning`	`warn` [7]	Warning conditions	
5	Notice	`notice`		Normal but significant conditions	Conditions that are not error conditions, but that may require special handling.[8]
6	Informational	`info`		Informational messages	
7	Debug	`debug`		Debug-level messages	Messages that contain information normally of use only when debugging a program.[8]

The meaning of severity levels other than *Emergency* and *Debug* are relative to the application. (Wikipedia)

inetd

- "super-server" daemon that manages other daemons
  - provides internet services
  - listens for traffic on designated ports
- Consults /etc/inetd.conf to determine which network ports to listen to
- Uses /etc/services or portmap daemon to map service names to port numbers

inetd.conf

- 1 service name
- 2 type of socket service uses (stream or dgram)
- 3 communication protocol uses (tcp or udp)
- 4 wait: service can process multiple request at one time

nowait – fork a new copy of daemon for each request

- 5 username under which daemon should run
- 6 fully qualified pathname of the daemon and its command-line arguments

vmstat

- Report on virtual memory usage

- Two arguments
  - number of seconds between measuring and reporting values
  - number of times to measure and report values

- Report
  - First line is the average values since last system reboot
  - Other lines represents current values

- man vmstat

netstat

- Report on network connection information

- Common Options
  - netstat –v
  - netstat –nt
  - netstat –nl
  - netstat –l
  - netstat –rn
  - netstat –s

- More Options